

# Structured Amortizing Bonds

Fixed Income Amortizing-Stepup-Sinking-Bond — Generated: 2026-09-11 14:27 UTC

## Inputs

| Field                              | Value                       |
|------------------------------------|-----------------------------|
| Model                              | Structured Amortizing Bonds |
| Settlement / Value Date            | 2026-08-30                  |
| Dated Date / Accrual Start         | 2026-06-20                  |
| First Coupon Date                  | 2026-12-20                  |
| Penultimate Coupon Date            | 2040-12-20                  |
| Maturity / Terminating Date        | 2041-06-20                  |
| Notional                           | 1000000                     |
| Coupon Rate                        | 0.0500                      |
| Pricing Basis                      | yield                       |
| Yield to Maturity                  | 0.0600                      |
| Market Clean Price (% of Par)      | 100.00                      |
| Principal Schedule Type            | straight_line               |
| Cashflow Schedule                  | 30 payment rows             |
| Coupon Frequency (Payments / Year) | 2                           |
| Day Count Convention               | ACT/ACT ISMA                |
| Scenario Shock (bp)                | 25                          |

## Results

| Metric                  | Value        |
|-------------------------|--------------|
| Fair Value (Clean)      | \$943,393.01 |
| Accrued Interest        | \$9,699.45   |
| Fair Value + Accrued    | \$953,092.46 |
| Yield to Maturity       | 6.0000%      |
| Duration                | 5.8385       |
| Modified Duration       | 5.6684       |
| BPV (+1bp Price Change) | -\$540.25    |
| Convexity               | 50.7291      |

## Analysis

The latest run includes parallel-rate clean-value scenarios, principal runoff, and the contractual coupon profile by payment date.

## Methodology

The engine generates deterministic coupon, principal and fixed-payment cashflows from the contractual schedule. Each cashflow is discounted using either the supplied yield to maturity or the interpolated zero curve. Accrued interest is separated from dirty value to produce clean value, and duration, modified duration, convexity and BPV are calculated consistently from the same cashflows and pricing basis.

| Stage                 | Implementation                                                                                                        |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Schedule construction | Generate contractual payment dates and apply the coupon and principal terms in effect for each period.                |
| Cashflow calculation  | Calculate accrual, coupon, principal redemption and any additional fixed payment, then roll opening notional forward. |
| Valuation             | Discount each cashflow using the selected yield or continuously compounded zero curve and separate accrued interest.  |
| Risk diagnostics      | Reprice under parallel shocks and calculate yield, duration, modified duration, convexity, BPV and principal runoff.  |

**References:** docs/methodology/amortizing\_stepup\_sinking\_bond.md, Math/GenericBonds.html, Math/lcb.html, Math/BondTable.html, Math/Daycount.html, Math/Dategen.html

## Testing Scope

| Test Area             | Coverage                                                                                                  |
|-----------------------|-----------------------------------------------------------------------------------------------------------|
| Contract dates        | Settlement, dated date, first and penultimate coupon dates, maturity and stub-period generation.          |
| Cashflow schedules    | Bullet, straight-line and custom sinking principal; level, step-up and step-down coupons; fixed payments. |
| Valuation             | Yield-based and curve-based discounting, accrued interest, clean/dirty reconciliation and yield solving.  |
| Risk outputs          | Parallel-rate scenarios, duration, modified duration, convexity, BPV and principal runoff.                |
| Independent benchmark | Published GenericBonds bullet and straight-line examples under the documented default terms.              |

## Testing Results

| Check                      | Observed                                                     | Acceptance Criterion                                                  | Status |
|----------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|--------|
| Cashflow identity          | Maximum residual 0.000e+00                                   | Coupon + principal + fixed payment = total cashflow                   | Pass   |
| Principal roll-forward     | Maximum link error 0.000e+00;<br>terminal residual 0.000e+00 | Consecutive notionals reconcile and principal is fully repaid         | Pass   |
| Clean/dirty reconciliation | Residual 0.000e+00                                           | Clean value + accrued interest = dirty value                          | Pass   |
| Parallel-rate direction    | Up-rate PV <= base PV <= down-rate PV                        | Positive-rate shock lowers value;<br>negative-rate shock raises value | Pass   |

Internal checks are calculated from this saved pricing run. The external comparison is included only when the saved dates, notional, coupon, frequency, day count, yield and principal style match the documented benchmark.

## External Benchmark Comparison

External GenericBonds benchmark - straight-line amortization

| Metric                  | DerivaPro    | External     | Difference | Status     |
|-------------------------|--------------|--------------|------------|------------|
| Fair Value (Clean)      | \$943,393.01 | \$943,393.01 | +0.000292  | Reconciled |
| Accrued Interest        | \$9,699.45   | \$9,699.45   | +0.003552  | Reconciled |
| Fair Value + Accrued    | \$953,092.46 | \$953,092.46 | +0.003844  | Reconciled |
| Duration                | 5.8385       | 5.8385       | -0.000019  | Reconciled |
| Modified Duration       | 5.6684       | 5.6684       | +0.000029  | Reconciled |
| BPV (+1bp Price Change) | \$-540.25    | \$-540.25    | -0.003649  | Reconciled |
| Convexity               | 50.7291      | 50.7291      | -0.000009  | Reconciled |

## Limitations

This report reflects a single pricing run under the stated assumptions. Model outputs should be independently reviewed before use in production or commercial decision-making.